STA 325 Case Study

```
library(tidyverse)
library(knitr)

train_path <- "data/data-train.csv"
raw <- read_csv(train_path, na = c("", "NA"), show_col_types = FALSE)
```

Data Processing

We see that the existing data contains Re and Fr values that are collected at distinct levels with low cardinality, instead of continuous values. It includes a valid infinity value for one of the predictors that is not a result of missing data. We compute standard deviation, skewness, and kurtosis from raw moments.

```
# predictor value distribution
level_table <- function(x, nm) {
  tibble(level = as.character(x)) |>
    count(level, name = "n") |>
    mutate(predictor = nm, .before = 1)
}
predictor_levels <- bind_rows(level_table(raw$Re, "Re"),
                              level_table(raw$Fr, "Fr"))
predictor_levels |>
  kable()
```

predictor	level	n
Re	224	39
Re	398	19
Re	90	31
Fr	0.052	36
Fr	0.3	23

predictor	level	n
Fr	Inf	30

```
# compute central moments + summary stats
m1 <- raw$R_moment_1
m2 <- raw$R_moment_2
m3 <- raw$R_moment_3
m4 <- raw$R_moment_4

analysis <- raw |>
  mutate(
    variance = m2 - m1^2,
    mean = m1,
    sd = sqrt(variance),
    central_moment_3 = m3 - 3 * m2 * m1 + 2 * m1^3,
    central_moment_4 = m4 - 4 * m3 * m1 + 6 * m2 * m1^2 - 3 * m1^4,
    skewness = central_moment_3 / sd^3,
    kurtosis = central_moment_4 / sd^4
  )
```

```
response_names <- c("mean", "sd", "skewness", "kurtosis")
response_summary <- analysis |>
  summarise(across(all_of(response_names),
                   list(min = min, q1 = ~ quantile(.x, .25), median = median,
                        mean = mean, q3 = ~ quantile(.x, .75), max = max,
                        sd = sd))) |>
  pivot_longer(everything(), names_to = c("response", ".value"), names_pattern = "(.*)_(min|
response_summary |>
  kable(digits = 2)
```

response	min	q1	median	mean	q3	max	sd
mean	0.00	0.00	0.00	0.04	0.09	0.17	0.06
sd	0.01	0.16	0.28	3.65	0.73	32.32	8.95
skewness	11.97	72.55	110.12	162.81	269.54	344.91	113.10
kurtosis	150.53	5622.34	12158.72	39749.56	72732.45	132136.73	38100.12

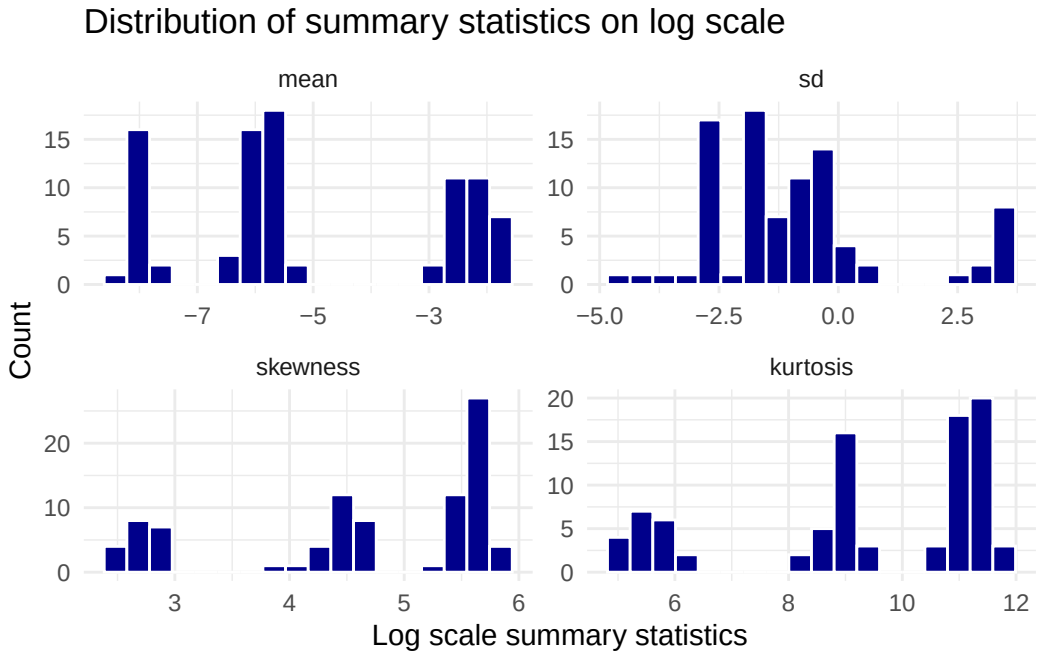
Exploratory data analysis

```
predictor_dist <- analysis |>
  mutate(Fr_condition = if_else(is.infinite(Fr), "Inf", as.character(Fr))) |>
  group_by(Re, Fr_condition) |>
  summarise(n = n(),
            St_min = min(St),
            St_max = max(St),
            St_levels = n_distinct(St), .groups = "drop")
predictor_dist |>
  kable()
```

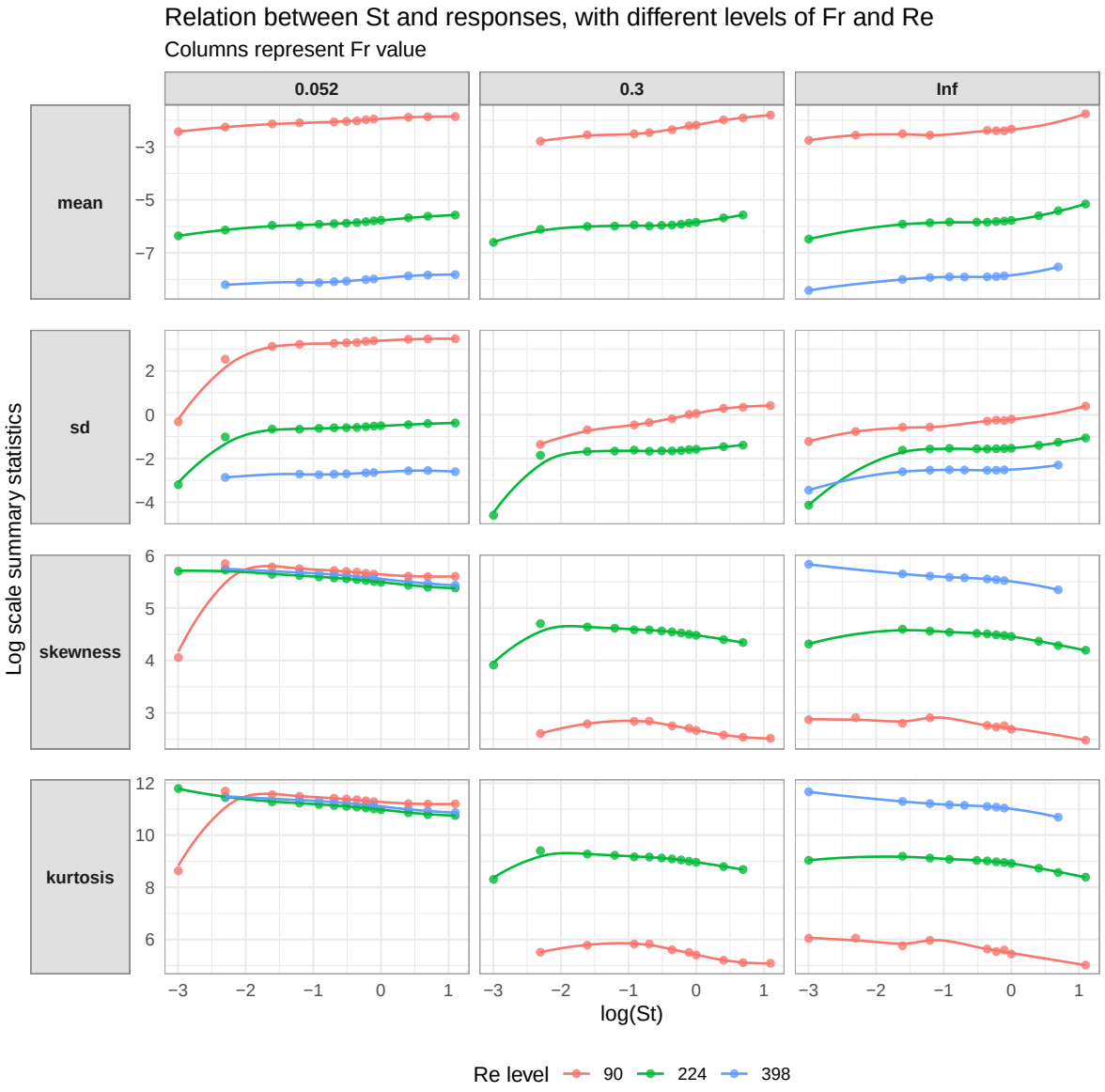
Re	Fr_condition	n	St_min	St_max	St_levels
90	0.052	12	0.05	3	12
90	0.3	10	0.10	3	10
90	Inf	9	0.05	3	9
224	0.052	14	0.05	3	14
224	0.3	13	0.05	2	13
224	Inf	12	0.05	3	12
398	0.052	10	0.10	3	10
398	Inf	9	0.05	2	9

```
response_long <- analysis |>
  select(Re, Fr, St, all_of(response_names)) |>
  pivot_longer(all_of(response_names), names_to = "response", values_to = "value") |>
  mutate(Fr_condition = if_else(is.infinite(Fr), "Inf", as.character(Fr)),
         log_value = log(value),
         log_St = log(St),
         response = fct_relevel(response, "mean", "sd", "skewness", "kurtosis"))

ggplot(response_long, aes(log_value)) +
  geom_histogram(bins = 18, color = "white", fill = "darkblue") +
  facet_wrap(~ response, scales = "free") +
  labs(x = "Log scale summary statistics",
       y = "Count",
       title = "Distribution of summary statistics on log scale") +
  theme_minimal()
```




```
ggplot(response_long, aes(log_St, log_value, color = factor(Re))) +
  geom_point(alpha = .8) +
  geom_smooth(method = "loess", formula = y ~ x, se = FALSE, linewidth = .6) +
  facet_grid(rows = vars(response), cols = vars(Fr_condition), scales = "free_y",
             switch = "y") +
  labs(x = "log(St)",
       y = "Log scale summary statistics",
       color = "Re level",
       title = "Relation between St and responses, with different levels of Fr and Re",
       subtitle = "Columns represent Fr value") +
  theme_minimal() +
  theme(
    panel.border = element_rect(color = "grey65", fill = NA, linewidth = .45),
    panel.spacing.y = grid::unit(1.1, "lines"),
    strip.placement = "outside",
    strip.background = element_rect(fill = "grey88", color = "grey55"),
    strip.text.y.left = element_text(face = "bold", angle = 0),
    strip.text.x = element_text(face = "bold"),
    legend.position = "bottom"
  )
```


Relation between St and responses, with different levels of Fr and Re
Columns represent Fr value
0.052
0.3
Inf
mean
sd
skewness
kurtosis
Log scale summary statistics
log(St)
Re level
90
224
398